A PROJECT TITLE

ON

RARE DISEASE DETECTION USING DBSCAN-BASED ANOMALY DETECTION

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING

BACHELOR OF TECHNOLOGY

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by

NAME : KODEBOINA PRIYADARSHINI ROLL NO: 2520030067

NAME : SAMBAVARAM SARVESWARA REDDY ROLL NO: 2520030049

NAME: JAMPANI POORNESH KUMAR ROLL NO:- 2520030055

Under the Esteemed guidance of

G.RAVINDRA BABU

Assistant Professor, CSE

DR.NIRMALA JYOTHI

Associate Professor, CSE

CSE

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)
Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043
Phone No. 7815926816, www.klh.edu.in

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

Name of Title: Rare Disease Detection Using DBSCAN-Based Anomaly Detection

1.ABSTRACT:

Rare diseases are very hard to detect because they occur in very few people and their symptoms often look similar to common illnesses. Traditional diagnosis methods usually depend on large sets of labelled medical data. But for rare diseases, such labelled data is not available, which makes detection even more challenging. To solve this, we can use DBSCAN (Density-Based Spatial Clustering of Applications with Noise), a clustering algorithm. DBSCAN groups patients with similar health data into clusters. If a patient's data does not fit into any cluster, they are marked as an *outlier*. These outliers may represent people with rare diseases or unusual health conditions.

The main advantage of DBSCAN is that it does not require us to decide the number of clusters beforehand. This is very useful in healthcare, where the number of disease types is unknown. DBSCAN can also detect clusters of different shapes and sizes, which makes it suitable for complex medical data such as lab results, genetic information, or clinical records. However, using DBSCAN requires careful preparation. Medical data must be cleaned and normalized so that no single feature dominates the clustering.

In conclusion, DBSCAN-based anomaly detection is a promising way to identify rare diseases early. By highlighting unusual patient data, doctors can investigate further and improve diagnosis. While it is not a complete solution, it provides a strong tool to support medical experts in detecting rare conditions that are otherwise overlooked.

2.INTRODUCTION:

Rare diseases pose a major challenge in healthcare because they affect only a small portion of the population and often mimic the symptoms of more common illnesses. This makes them difficult to diagnose using traditional methods, which usually depend on large amounts of labelled medical data. Unfortunately, such data is scarce for rare conditions, leaving doctors with limited resources to identify them early. As a result, many patients experience delayed diagnosis or misdiagnosis, which can impact treatment outcomes. To address this issue, researchers are turning to advanced data-driven techniques like DBSCAN. DBSCAN is a clustering algorithm that groups patients with similar health records together and identifies those who do not fit into any group as *outliers*. These outliers may represent individuals with rare diseases or unusual health patterns. By detecting anomalies in complex datasets such as genetic information, lab results, or clinical histories, DBSCAN offers a promising way to highlight potential rare cases for further medical investigation.

3.SOFTWARE REQUIREMENTS:

The following software is used to develop this project:

- **Operating System:** Windows 10 / Ubuntu (Linux)
- **IDE / Notebook:** Jupyter Notebook, VS Code, or PyCharm
- **Scikit-learn** → DBSCAN implementation, anomaly detection, clustering
- **NumPy** → numerical computations and matrix operations
- **Pandas** → data pre-processing, cleaning, and handling patient datasets
- **Matplotlib / Seaborn** → visualization of clusters, anomalies, and patient profile.

Signature of the Guide

Course Instructor